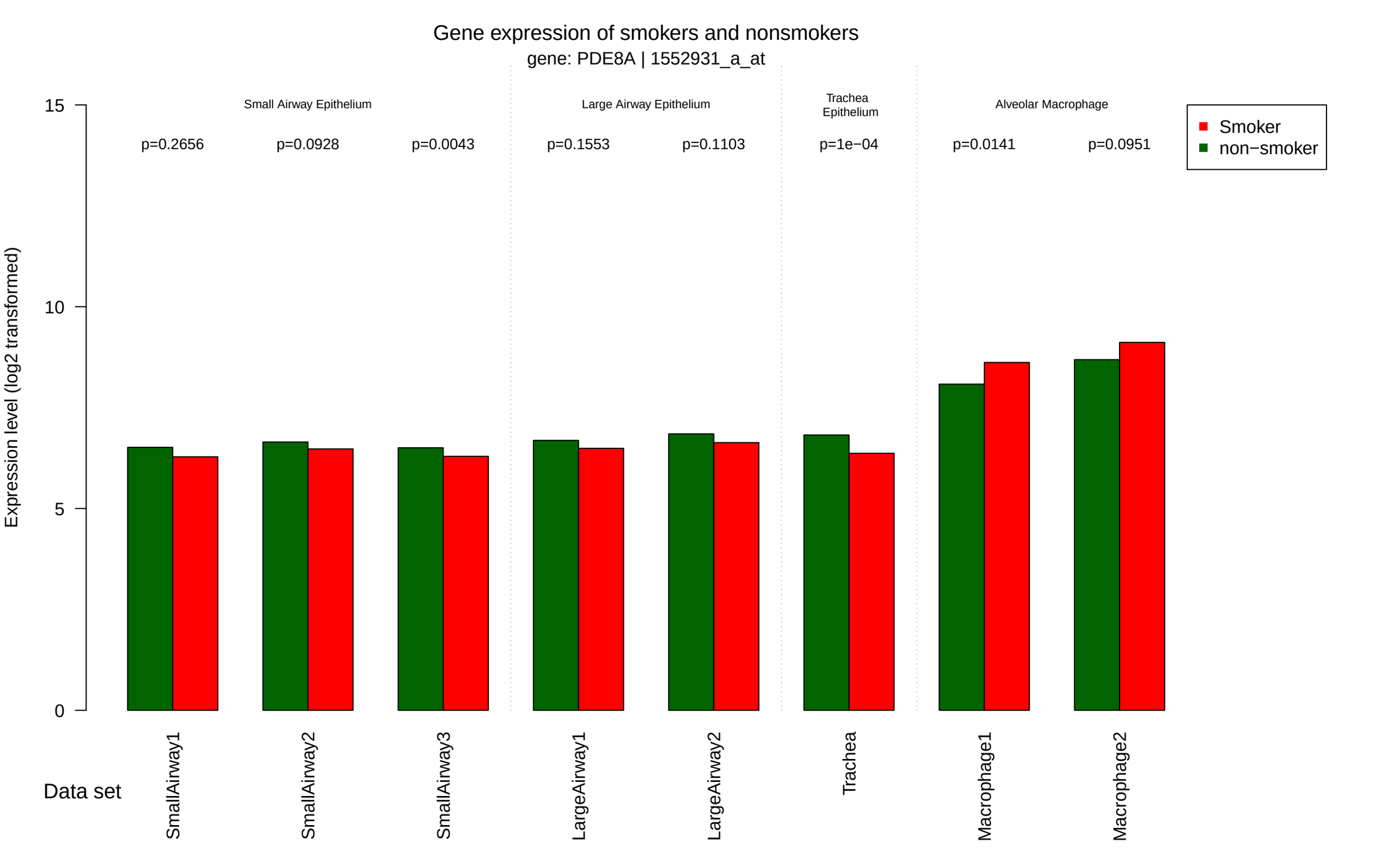
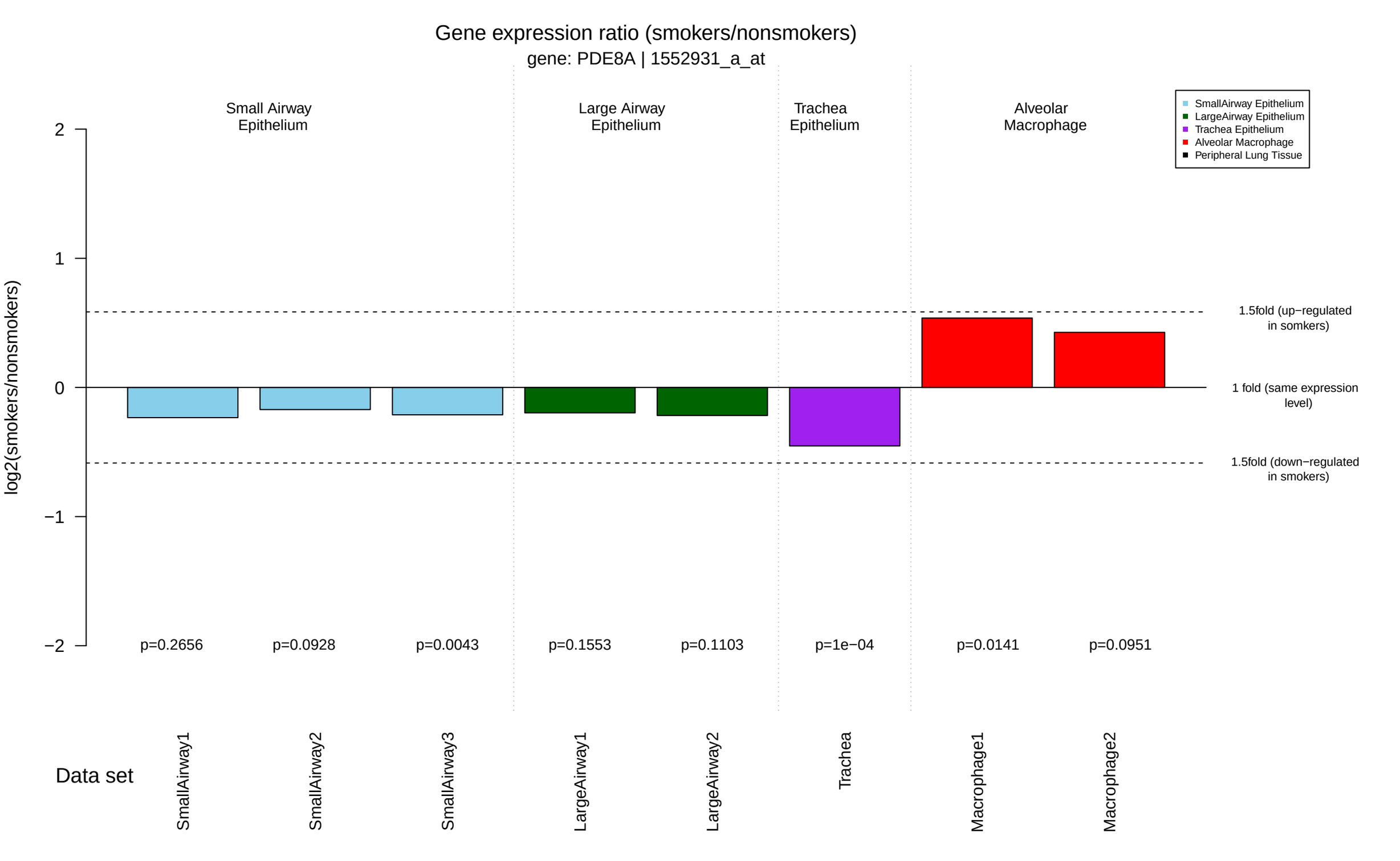
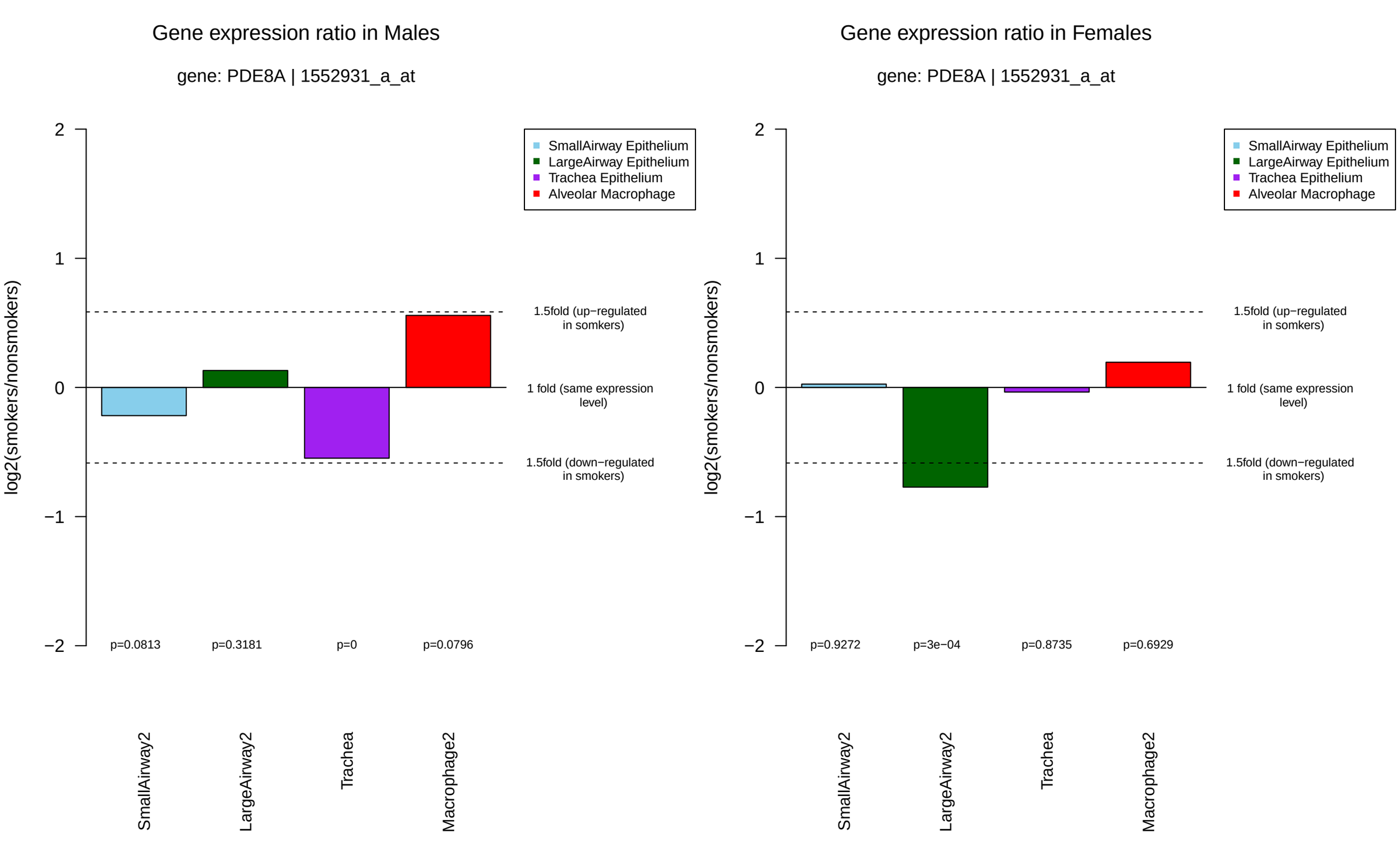
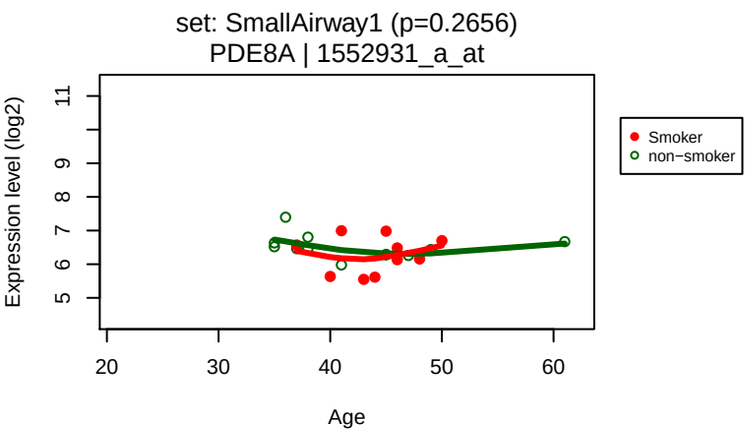
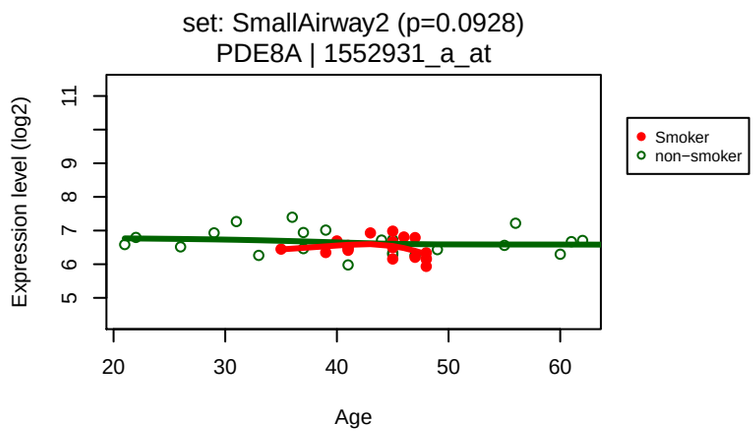
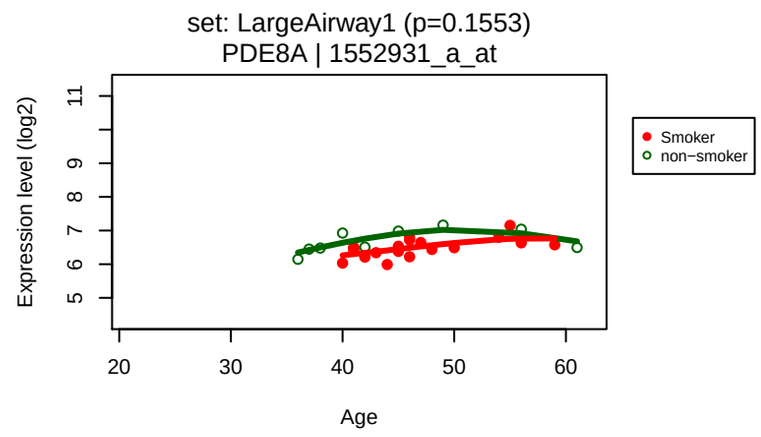

Gene expression of smokers and nonsmokers
gene: PDE8A | 1552931_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.2656
p=0.0928
p=0.0043
p=0.1553
p=0.1103
p=1e−04
p=0.0141
p=0.0951
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

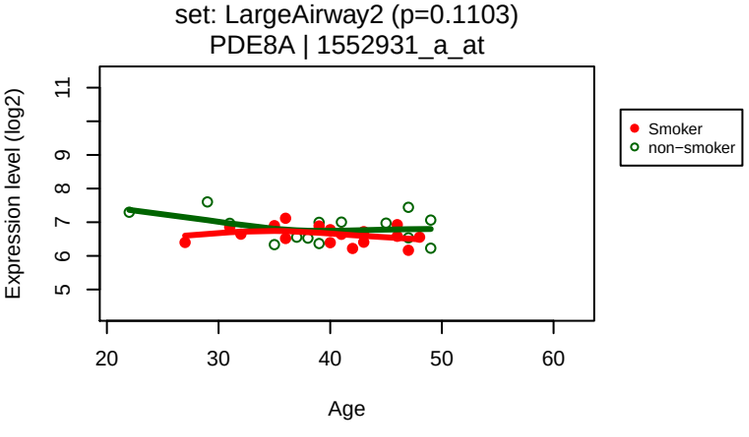

Gene expression ratio (smokers/nonsmokers)
gene: PDE8A | 1552931_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2656
p=0.0928
p=0.0043
p=0.1553
p=0.1103
p=1e−04
p=0.0141
p=0.0951
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

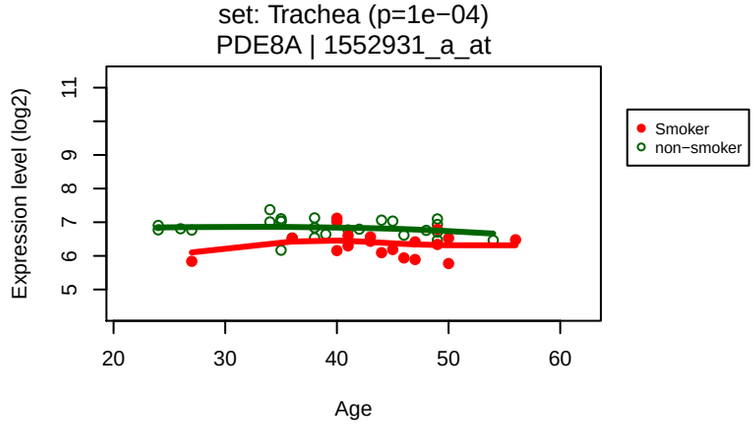

Gene expression ratio in Males
gene: PDE8A | 1552931_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0813
p=0.3181
p=0
p=0.0796
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: PDE8A | 1552931_a_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9272
p=3e−04
p=0.8735
p=0.6929
SmallAirway2
LargeAirway2
Trachea
Macrophage2

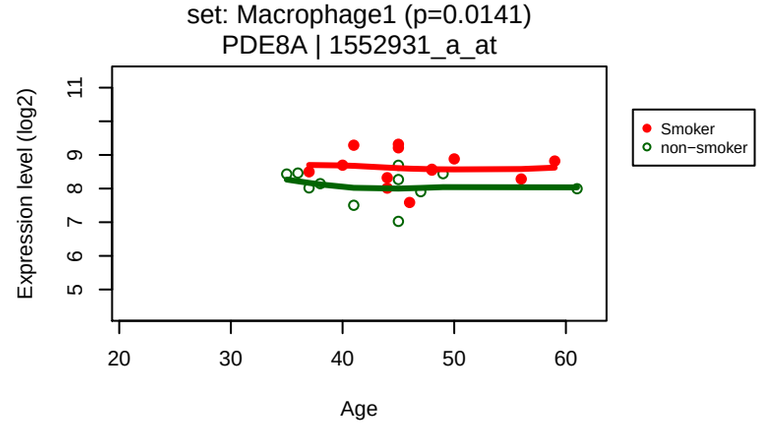

set: SmallAirway1 (p=0.2656)
PDE8A | 1552931_a_at
Expression level (log2)
Age
Smoker
non−smoker

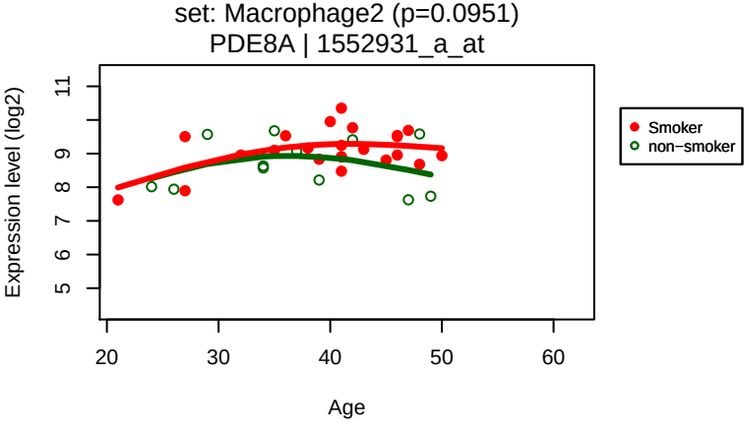

set: SmallAirway2 (p=0.0928)
PDE8A | 1552931_a_at
Expression level (log2)
Age
Smoker
non−smoker

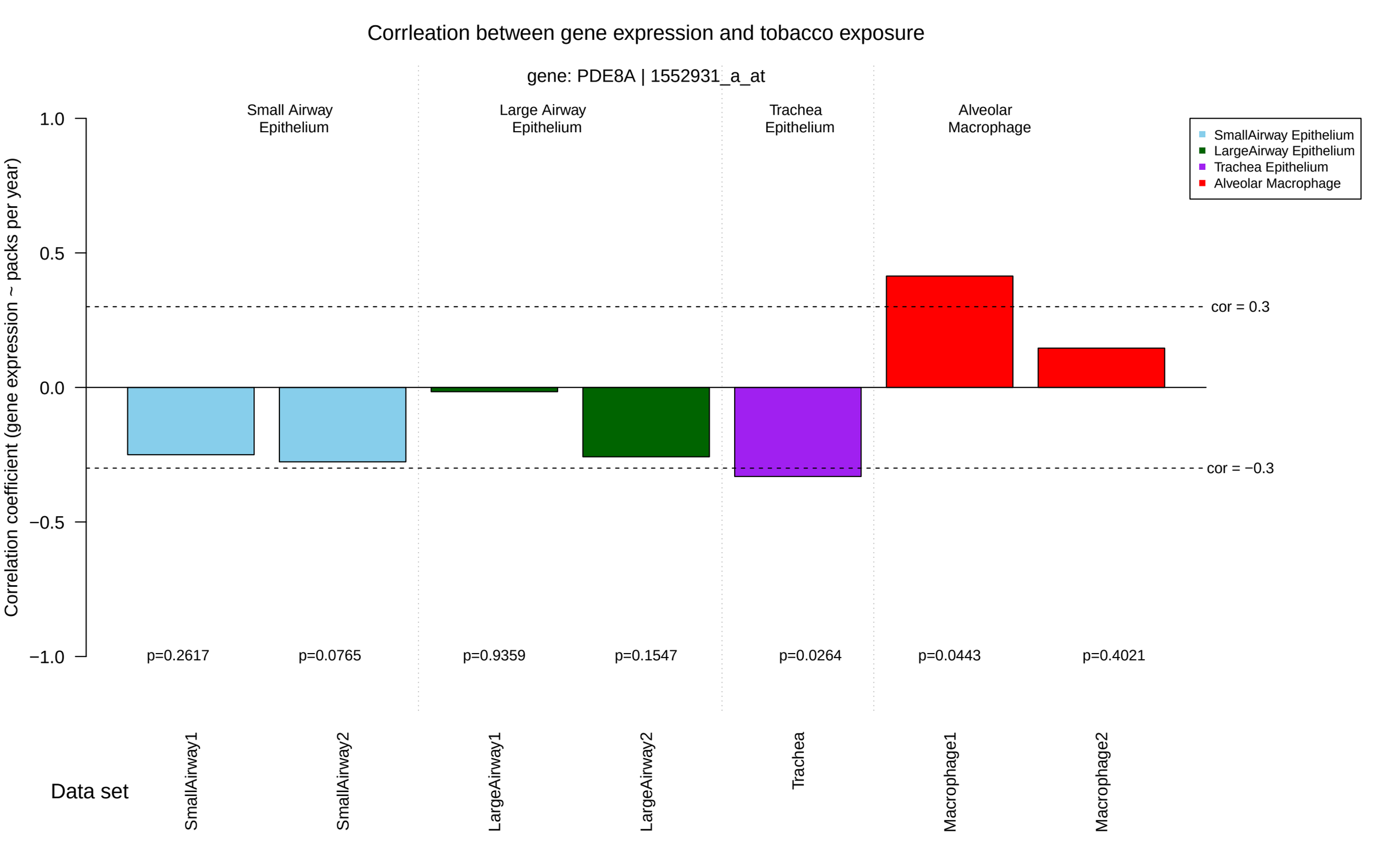

set: LargeAirway1 (p=0.1553)
PDE8A | 1552931_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.1103)
PDE8A | 1552931_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=1e−04)
PDE8A | 1552931_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0141)
PDE8A | 1552931_a_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0951)
PDE8A | 1552931_a_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: PDE8A | 1552931_a_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2617
p=0.0765
p=0.9359
p=0.1547
p=0.0264
p=0.0443
p=0.4021
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2